

Lexical Grammar

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1 Regex Substitutions

$$\begin{array}{lll}
L & := & [\mathbf{a} - \mathbf{z}] \quad (1) \\
U & := & [\mathbf{A} - \mathbf{Z}] \quad (2) \\
D & := & [0 - 9] \quad (3) \\
H & := & [\mathbf{a} - \mathbf{fA} - \mathbf{F0} - 9] \quad (4) \\
O & := & [0 - 7] \quad (5) \\
B & := & [01] \quad (6)
\end{array}$$

2 Identifiers, Constants, and Comments

$_ \{0, 2\}L(? : L U D _)^*$	$\rightarrow$	IDENTIFIER	(7)
$U(? : L U D)^*$	$\rightarrow$	TYPE_IDENTIFIER	(8)
$[\backslash + -]?D+$	$\rightarrow$	NUMERIC_CONST Integer	(9)
$0[xX]H\{1, 16\}$	$\rightarrow$	NUMERIC_CONST Hex	(10)
$0[oO]O\{1, 32\}$	$\rightarrow$	NUMERIC_CONST Octal	(11)
$0[bB]B\{1, 64\}$	$\rightarrow$	NUMERIC_CONST Binary	(12)
$[\backslash + -]?D + [eE][\backslash + -]?D+$	$\rightarrow$	NUMERIC_CONST Scientific Notation	(13)
$[\backslash + -]?((? : D + (? : [_]D^*)[.]D+)$	$\rightarrow$	NUMERIC_CONST Float	(14)
$r?"((? : [\ ^ "\\\\" \\")^*)"$	$\rightarrow$	STRING_LITERAL	(15)
$\#.\ * \backslash n\$$	$\rightarrow$	COMMENT	(16)

3 Symbols

,	→	COMMA	(17)
;	→	SEMI	(18)
(	→	LEFT_PAREN	(19)
)	→	RIGHT_PAREN	(20)
[	→	LEFT_SQUARE	(21)
]	→	RIGHT_SQUARE	(22)
{	→	LEFT_CURLY	(23)
}	→	RIGHT_CURLY	(24)

4 Keywords

and	→	LOGICAL_AND	(25)
bool	→	BOOL	(26)
break	→	BREAK	(27)
class	→	CLASS	(28)
elif	→	ELIF	(29)
else	→	ELSE	(30)
enum	→	ENUM	(31)
exit	→	EXIT	(32)
false	→	FALSE	(33)
float32	→	FLOAT32	(34)
float64	→	FLOAT64	(35)
for	→	FOR	(36)
func	→	FUNCTION	(37)
if	→	IF	(38)
iface	→	INTERFACE	(39)
in	→	IN	(40)
int16	→	INT16	(41)
int32	→	INT32	(42)
int64	→	INT64	(43)
int8	→	INT8	(44)
mut	→	MUTABLE	(45)
or	→	LOGICAL_OR	(46)
private	→	PRIVATE	(47)
public	→	PUBLIC	(48)
string	→	STRING	(49)
true	→	TRUE	(50)
uint16	→	UINT16	(51)
uint32	→	UINT32	(52)
uint64	→	UINT64	(53)
uint8	→	UINT8	(54)
while	→	WHILE	(55)

5 Operators

$\wedge\wedge=$	→	EXPONENT_ASSIGN	(56)
$\wedge\wedge$	→	EXPONENT	(57)
$\wedge=$	→	XOR_ASSIGN	(58)
$\wedge$	→	XOR	(59)
$\%=$	→	MOD_ASSIGN	(60)
$\%$	→	MOD	(61)
$\ast=$	→	MUL_ASSIGN	(62)
$\ast$	→	MUL	(63)
$//=$	→	INTEGER_DIV_ASSIGN	(64)
$//$	→	INTEGER_DIV	(65)
$/=$	→	DIV_ASSIGN	(66)
$/$	→	DIV	(67)
$++$	→	INC	(68)
$+=$	→	ADD_ASSIGN	(69)
$+$	→	ADD	(70)
$--$	→	DEC	(71)
$-=$	→	SUB_ASSIGN	(72)
$-$	→	SUB	(73)
$<<$	→	SHIFT_LEFT	(74)
$<=$	→	LE_OP	(75)
$<$	→	LT_OP	(76)
$>>$	→	SHIFT_RIGHT	(77)
$>=$	→	GE_OP	(78)
$>$	→	GT_OP	(79)
$!=$	→	NE_OP	(80)
$==$	→	EQ_OP	(81)
$=$	→	ASSIGN	(82)
$\&=$	→	AND_ASSIGN	(83)
$\&$	→	AND	(84)
$ =$	→	OR_ASSIGN	(85)
$ $	→	OR	(86)
$?$	→	TERNARY	(87)
$.$	→	DOT	(88)